FARAAZ MOHAMMED

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Full Australian working rights (Temporary Graduate visa, subclass 485) — no sponsorship required • Available immediately

PROFILE

Master of Artificial Intelligence graduate (RMIT, 2026) with a B.Tech in IT and an AI/ML engineering internship shipping production pipelines at national scale. Builds automations and AI tooling others can reuse — MCP servers, Claude Code skills, API routers — and scopes problems with non-technical stakeholders before writing code. Uses AI where it earns its place, and conventional software when it is the better call. Seeking an AI Solutions Engineer role at pay.com.au, Melbourne.

TECHNICAL SKILLS

Languages: Python (primary), TypeScript, JavaScript, SQL, Bash, Kotlin

Automation & APIs: REST/JSON, Node.js, MCP servers, Kestra, Next.js, Prisma/Postgres, agent-style workflows

AI tools: Claude Code (skills, hooks, subagents), Cursor, Anthropic API (prompt caching, routing), Ollama, LLM fine-tuning (Unsloth, QLoRA, GGUF), prompt engineering, retrieval-grounded generation

Engineering habits: evaluation / test harnesses (IoU, sensitivity analysis, JUnit), Git / GitHub, Docker (basic), Linux, Vercel

Also: scikit-learn, XGBoost, OpenCV, GeoPandas (pipeline context)

EXPERIENCE

AI/ML Engineering Intern — Prodegee *Mar 2026 – Jun 2026*
Remote, Melbourne · RMIT industry capstone, Team DS4 — building-footprint inference for off-grid electrification planning

- Designed and built the school-geolocation algorithm at the core of the team's pipeline (7.2M Overture building polygons, BFS spatial clustering, composite scoring); ran it on 18,121 schools in Côte d'Ivoire and ported it to Burundi and Chad.
- Built an IoU evaluation harness for compound polygons and traced all three generators' failures to a ~35× over-scoped upstream cluster rather than a fitting fault, which redirected the next assignment.
- Built a ±20% sensitivity harness that identified cluster radius as the dominant parameter (±36–44% swing across 191 villages), driving a 75 → 200 m recalibration that fixed a 90%+ household under-count (0.069 → 0.415).
- Co-built a 40-school satellite-verified ground-truth set and used it to correct a published baseline (6–13% → 37%); published a negative result showing a proximity penalty hurt accuracy at every threshold.
- Led 3 of 10 team assignments and was primary author of 3 of 4 formal deliverables (68 commits, 7 peer reviews); pinned upstream inputs by commit SHA for reproducible re-runs and wrote the semester-2 handoff memo.
- Built a Claude API model router that classifies task complexity with a cheap Haiku call and routes to Haiku / Sonnet / Opus, keeping cost proportional to difficulty.
- Replaced a binary distance cutoff with exponential-decay scoring and a weighted confidence score; added per-country sigmoid normalisation so one threshold works in all three countries; authored cross-team handoff contracts.

Tools: Python, Kestra, Git, Claude Code (MCP, hooks), scikit-learn, pandas, GeoPandas, Shapely, Docker (basic)

Dispatch Team Member — Amart Furniture *Feb 2025 – Mar 2026*
Melbourne, VIC · part-time, on-site, concurrent with full-time study

- Tracked and reconciled stock in the inventory system, processed customer orders and resolved fulfilment discrepancies across 5 Melbourne stores; trained a new team member.

EDUCATION

Master of Artificial Intelligence — RMIT University *Jul 2024 – Oct 2026*
Melbourne, VIC · Coursework: AI Systems Design, Machine Learning, Computer Vision, Intelligent Decision Making (ASP/Clingo)

Bachelor of Technology (Information Technology) — Keshav Memorial Institute of Technology *Dec 2020 – May 2024*
Hyderabad, India (affiliated to JNTUH) · Coursework incl. Machine Learning, Neural Networks & Deep Learning, Data Mining

KEY PROJECTS

Gemma Delegation System — local/cloud LLM hybrid · Node.js, MCP SDK, Ollama, Unsloth/QLoRA *May 2026*

- Built an MCP server that lets Claude Code or Cursor delegate lightweight tasks to a locally hosted Gemma model, failing open to the frontier model whenever Ollama is unavailable so delegation never blocks the user.
- Closed the loop: accept/reject feedback updates a capability profile and curates a JSONL dataset; a Colab notebook fine-tunes on confirmed examples (4-bit QLoRA, LoRA r=16) and exports GGUF back into Ollama.

  ↳ GitHub: github.com/faraaz817/gemma-delegation-system

Agent skills for Claude Code / Cursor · skills, hooks, reusable workflows *2026*

- Authored five reusable skills — CO-STAR requirements builder, Socratic tutor, better-way reviewer, skill sync, and a pikepdf CLI that unlocks signature-locked PDFs — packaged for others to pick up and run.

  ↳ GitHub: github.com/faraaz817/My_Skills

MGM Prayer Alerts — Android app for Melbourne Grand Mosque · Kotlin, AlarmManager, JUnit *Aug 2026*

- Wrote functional requirements (scope, decision log, acceptance criteria) with stakeholders before coding; shipped an app now in use by community members.
- Fixed a critical AM/PM storage bug and a refresh alarm that stopped re-arming; unit-tested time parsing.
  ↳ GitHub: github.com/faraaz817/MgmPrayerAlerts

Bro — interactive AI portfolio chatbot · Next.js, TypeScript, Prisma/Postgres, Anthropic API *2026*

- Recruiter-facing site that answers questions from a profile corpus via server-side retrieval with anti-hallucination constraints; Anthropic API, deployed on Vercel.
  ↳ Live: job-go-mauve.vercel.app/portfolio

Time Flow — ADHD-focused planner (Android, closed source) · JavaScript, Kotlin *Jul 2026 – present*

- Building an Android planner for commercial release (vanilla-JS UI in a Kotlin shell); set up CI that runs the Kotlin and JavaScript test suites on every PR and publishes signed, apksigner-checked releases.
  ↳ Closed source (in development for commercial release) · demo on request
- **Fire Detection & Alerting System (B.Tech capstone, team of 4):** trained a custom OpenCV Haar cascade (Python) for real-time fire detection, chosen over heavier detectors for latency; a threaded pipeline raises an alarm and sends WhatsApp alerts to the nearest fire station.

PUBLICATION

K. M. Asudaria, **M. F. Abdul Khadeer**, S. S. Rafai, K. S. Reddy. "A Comparative Study of Edge Detection Techniques to Identify Maize Leaf Diseases using Machine Learning." *Quest Journal of Software Engineering and Simulation*, Vol. 9(12), pp. 43–51, Dec 2023 (ISSN 2321-3795, peer-reviewed). Best result Sobel + MLP at 94.75%. Implemented the Canny arm (OpenCV + scikit-learn). ↳ questjournals.org/jses

CERTIFICATIONS, AWARDS & LEADERSHIP

- Head, Vachan Speakers Club, KMIT (2022–23) — led public-speaking workshops, debates and communication seminars.
- IBM Z Xplore – Concepts and IBM Z & LinuxONE Community Contributor – Level 1 (IBM digital badges via Credly, 2026).
- Smart India Hackathon delegate, Government of India (2022) · Certificate of Publication, Quest Journals (2023).
- Volunteer fundraiser, Taher Foundation (2023).

LANGUAGES

English (native / bilingual) • Urdu (native / bilingual) • Hindi (full professional) • Telugu (elementary)

REFEREES

Available on request.